Intelligent Property Management Workflow with OCR-Enabled Automated Billing Classification and Tenant Notification

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CHAPTER I

THE PROBLEM AND ITS BACKGROUND

This chapter presents the background and related literature of the study entitled “Intelligent Property Management Workflow with OCR-Enabled Automated Billing Classification and Tenant Notification.” It also presents the significance of the study, the statement of the problem, and the scope and limitations of the system.

1.1 Introduction

Property management involves maintaining tenant records, managing utility bills, computing charges, monitoring payment status, and providing billing information to tenants. These activities require the organized handling of property records, tenant information, billing documents, and other administrative records. Cammarata (2023) discusses the importance of records and information management in real-estate operations, while Teo et al. (2024) examine property-management functions and services. When property managers handle these activities through separate or manual methods, organizing information and maintaining consistent work processes can become more difficult.

To support these activities, digital approaches provide a way to centralize property information and organize regular management tasks. Property Technology (PropTech) refers to the use of digital technologies in real-estate and property-management

activities. Siniak et al. (2020) discuss the role of PropTech in changing real-estate operations and services, while Oyetunji et al. (2025) examine digital solutions for streamlining property-management operations. Ding and Sun (2024) review technology innovations in housing management, while Al-haimi et al. (2025) discuss the benefits and challenges of digital transformation in the real-estate industry. Taken together, these studies show how digital technologies support property and housing-management activities and provide a basis for using centralized systems to organize property information and management processes.

One important area in property management is billing. Utility billing requires information from billing documents before charges can be processed, including amounts, dates, billing periods, providers, and account or reference identifiers. Optical Character Recognition (OCR) provides a way to extract text and relevant information from supported documents such as receipts and invoices (Kumar et al., 2020). Saout et al. (2024) discuss invoice-data extraction as a process involving document digitization, OCR, information extraction, and recognition of document structures. Mohan (2023) also presents a staged pipeline for extracting key-value information from commercial invoices. Taken together, these studies support the use of OCR and document-processing methods to extract and organize information from billing documents for later processing.

After billing information is extracted, it still needs to be validated, classified, and processed according to defined billing conditions. After billing information is extracted, it still needs to be validated, classified, and processed according to defined billing

conditions. Rule-based processing uses predefined conditions and logical rules to support structured operational decisions and information handling. Baralis et al. (2022) discuss business-rule engines that represent and execute operational decisions in a structured way. In relation to process automation, Geyer-Klingeberg et al. (2024) present a rule-based method that analyzes decision logic and classifies business processes according to their suitability for robotic process automation. These ideas are relevant to a billing workflow in which validation, classification, calculation, and exception-handling conditions are defined in advance. These ideas are relevant to a billing workflow in which validation, classification, calculation, and exception-handling conditions are defined in advance.

Based on these ideas, the proposed system connects property management, OCR-based extraction of supported utility-bill documents, validation, rule-based billing classification, billing computation, exception handling, and Web Push Notifications within one web-based workflow. In this study, intelligent processing refers to rule-based processing that uses predefined and traceable conditions to validate billing entries, classify billing information, compute applicable charges, identify processing exceptions, and route records to the Manager for review when required. The study focuses on organizing rental and utility billing activities within a centralized system while applying defined rules to billing and exception-handling processes.

1.2 Review of Related Literature

Digital Property Management and PropTech

Digital property management increasingly relies on technology to organize property details, management activities, billing data, and related services. Several studies have examined this shift from different perspectives: Teo et al. (2024) discuss technology and digitalization in property-management functions and service delivery, while Ding and Sun (2024) review technology innovations in housing management. In this context, PropTech refers to the use of digital technologies in the real-estate and property-management sector. Siniak et al. (2020) examine the impact of PropTech on real-estate industry growth, Saputra et al. (2023) review digital-transformation trends in the property-management industry, and Al-haimi et al. (2025) synthesize the technologies, benefits, and challenges associated with digital transformation in real estate. In addition, Cammarata (2023) discusses the importance of records and information management in managing risks during real-estate changes. Taken together, these studies support the relevance of centralized digital tools for property information and administrative processes. Building on this literature, the present study proposes a centralized web-based workflow that organizes property, unit, tenant, utility, billing, payment, exception, and receipt information for selected boarding houses and residential apartments.

OCR and Billing Document Extraction

Within property-management operations, billing is one area where digital processing can support routine administrative work. Billing documents contain information such as amounts, dates, billing periods, providers, and account or reference identifiers. Kumar et al. (2020) examine the extraction of information from bill receipts using Optical Character Recognition (OCR), while Saout et al. (2024) provide an overview of invoice-data extraction involving document digitization, OCR, information extraction, and document-structure recognition. Similarly, Mohan (2023) presents an end-to-end pipeline for custom key-value extraction from commercial invoices. However, extraction is only the first stage of billing processing. In the proposed system, OCR is used to process supported standard digital PDFs and clear images, after which the extracted information undergoes validation before billing processing. Records that do not meet the required validation conditions are sent to the Exception Queue for Manager review. Thus, OCR is treated as a method for extracting information from billing documents rather than as a guarantee that every extracted value is accurate.

Rule-Based Billing Processing

Once billing information has been extracted, it must be checked and processed according to defined conditions. Rule-based processing uses explicit conditions and decision rules to classify, validate, and route information. Wu et al. (2022) demonstrate the use of rule-based information extraction to organize domain-specific information, while Baralis et al. (2022) discuss business-rule engines for representing and executing operational decisions in a structured manner. Extending this idea to process automation,

Geyer-Klingeberg et al. (2024) present a rule-based method that analyzes decision logic and classifies business processes according to their suitability for robotic process automation, while Kim et al. (2011) examine proactive exception handling in business processes. These studies support the use of predefined and traceable conditions to determine whether a record proceeds through normal processing or requires additional review. Applying these principles, the proposed system uses defined rules for billing validation, provider and account/reference mapping, duplicate detection, and identifying individual or shared billing arrangements. Records that do not meet the defined conditions are sent to the Exception Queue for Manager review. These rules, as well as the separation between Tenant Reported Paid and Manager-confirmed payment status, are system requirements rather than findings established by the cited studies. Digital receipts are created only after payment confirmation.

Shared Utility Billing and Cost Allocation

Shared utility billing presents a related issue because a single charge may need to be allocated among several occupants. Levinson and Niemann (2004) examine apartment energy use when landlords pay for utilities, while Mayr and Souza (2025) discuss coordination and incentive issues associated with shared heating costs. These studies provide context for understanding cost responsibility and consumption incentives in shared-utility arrangements. Nevertheless, they do not establish one universal allocation formula for rental properties or determine how the proposed system should calculate individual tenant charges. For this reason, the proposed system treats the

allocation method as a defined system requirement rather than as a formula established by the reviewed literature. For shared occupancy, the Manager configures the predefined allocation rule. The billing requirements also include reconciliation controls so that allocated charges equal the source billing amount. The specific allocation formula, rounding procedure, and reconciliation method are system-design decisions described and tested in the succeeding chapters.

Web Push Notifications and Secure Notification Delivery

After a billing record is finalized, tenants need to be informed that their billing information is available. The World Wide Web Consortium (2024a) describes the Push API, including push services and user subscriptions. The World Wide Web Consortium (2024b) also explains the role of service workers in handling web events in the background when they are included in the system implementation. Because Web Push involves browser subscriptions and authorization, security must also be considered. Carboneri et al. (2023) examine Web Push implementations in the wild and identify security concerns involving push subscriptions and authorization. In the proposed system, Web Push notifications serve as billing alerts that direct tenants to the authenticated web portal, where complete billing information is available. However, Web Push does not confirm payment, prove that payment occurred, or show whether the tenant read the notification. Tenant payment reporting therefore remains separate from Manager confirmation.



Iterative Software Development

Because the proposed system contains several connected functions, its development requires an organized and adaptable process. Iterative software development divides development into repeated cycles of building, reviewing, receiving feedback, and improving. Larman and Basili (2003) explain the development of iterative and incremental approaches, in which software is developed progressively through repeated cycles. Inayat et al. (2015) emphasize stakeholder involvement, communication, and requirements refinement in iterative requirements engineering, while Ramesh et al. (2021) identify challenges involving changing requirements, stakeholder participation, documentation, and communication in Agile software development. These studies support the use of Iterative Software Development Methodology in the present study. Specifically, the methodology organizes the proposed system into repeated cycles of requirements review, development, testing, feedback, and refinement. Although the literature supports this approach, the actual quality and performance of the proposed system are determined through implementation, testing, user feedback, and evaluation.

Software Quality Evaluation and Security

To evaluate the proposed system systematically, the study requires defined software-quality criteria. ISO/IEC 25010:2023 provides a product-quality model for systems and software quality requirements and evaluation (International Organization for Standardization, 2023). Guided by this model, the study evaluates five quality characteristics: Functional Suitability, Performance Efficiency, Interaction Capability,

Reliability, and Security. These characteristics provide a basis for examining whether the system performs its intended functions, performs efficiently, supports user interaction, operates consistently, and protects information. Security is particularly important because the system handles tenant and billing information. Republic Act No. 10173, or the Data Privacy Act of 2012, requires organizations that process personal information to use reasonable and appropriate measures to protect it (Republic of the Philippines, 2012). The National Privacy Commission (2016) further discusses safeguards for the confidentiality, integrity, and availability of personal data, while Zafar et al. (2022) review authorization and access-control strategies that restrict system functions and information according to user permissions. In the proposed system, role-based access separates Manager and Tenant functions. These sources guide the study's quality and security evaluation, but they do not prove that the system already meets the selected quality characteristics or privacy requirements. The Security evaluation is limited to the defined system functions, access controls, and security-related criteria included in the study's evaluation procedures. It does not constitute a legal audit, certification, or claim of complete compliance with the Philippine Data Privacy Act.

1.3 Significance of the Study

The study develops and evaluates a web-based property-management workflow that integrates electronic utility-bill intake, OCR-based information extraction, rule-based billing classification, tenant and billing record management, and Web Push

Notifications.th The study provides practical and academic value in the following areas. The primary beneficiaries are property managers and tenants, while future researchers and system developers may use the study as a reference for applying digital property-management technologies in small-scale residential settings.

In the global context, the study contributes an applied example of PropTech and digital property management by showing how existing technologies can be combined into one workflow for residential property operations. It also provides a local example of how digital property-management concepts can be applied in a smaller property-management setting. The study does not claim that the proposed workflow represents all property-management environments; rather, its significance lies in demonstrating how selected technologies may be integrated within the defined scope of the study.

In the economic context, the system is designed to reduce repetitive manual activities involved in receiving utility bills, recording billing information, calculating tenant charges, and communicating billing updates. Centralized records and predefined billing rules can also help property managers follow a more consistent billing process. However, actual reductions in operating costs, processing time, or late payments depend on the results of system testing and evaluation.

In the environmental context, the system uses electronic utility bills, digital billing records, and electronic notifications. This can reduce the need for paper-based

records during the billing process. The study does not claim a specific reduction in paper use because this is not directly measured.

In the societal context, the system provides property managers and tenants with organized billing information. Itemized billing records, defined billing rules, and automated notifications can help make tenant dues easier to understand and communicate. This can support clearer billing transactions between property managers and tenants. However, the study does not claim that the system will eliminate billing misunderstandings or disputes because those outcomes are not directly measured.

1.4 Statement of the Problem

This study develops and evaluates an Intelligent Property Management Workflow with OCR-Enabled Automated Billing Classification and Tenant Notification for selected boarding houses and residential apartments in Pasig City.

Specifically, the study seeks to answer the following questions:

1. What development stages are undertaken in developing the proposed system using the Iterative Software Development Methodology?
2. How are Optical Character Recognition (OCR), validation, Exception Queue handling, and rule-based billing classification integrated into the proposed system to process rental and utility billing information?

3. How does the developed system perform when evaluated using the selected ISO/IEC 25010:2023 quality characteristics in terms of:
    a. Functional Suitability;
    b. Performance Efficiency;
    c. Interaction Capability;
    d. Reliability; and
    e. Security?
4. What implementation guide and user manual can be prepared for the developed system?

1.5 Scope and Limitations of the Study

The study develops and evaluates a web-based property-management workflow entitled Intelligent Property Management Workflow with OCR-Enabled Automated Billing Classification and Tenant Notification. It aims to provide a centralized workflow for organizing property, unit, tenant, utility, rent, billing, payment, exception, receipt, and notification records. The system integrates electronic utility-bill intake, OCR-based document-information extraction, validation, deterministic rule-based billing classification, Exception Queue handling, Manager payment confirmation, digital receipt generation, and Web Push Notifications. In this study, the term intelligent refers to predefined,



explicit, deterministic, and traceable rules and does not refer to Artificial Intelligence, Machine Learning, Deep Learning, Predictive Analytics, Generative AI, or any machine-learning model.

The study covers selected boarding houses and residential apartments in Pasig City during Academic Year 2026–2027. The billing categories are limited to rent, electricity, and water. Rent is generated monthly according to the tenant's lease-anniversary date, is fixed and non-prorated, and includes one-month advance rent as a separate first-billing line item. The system supports Individual Billing Arrangement and Shared-Occupancy Billing Arrangement. Under shared occupancy, rent and applicable utility charges are allocated among active tenants according to a predefined rule selected or configured by the Manager. Allocated amounts use two decimal places, and the total of the allocated shares must equal the original source billing amount.

The system receives electronic utility bills through a designated property-management email account and uses OCR to extract available billing information from standard, digitally generated PDF files and sufficiently clear image files. The target fields include the amount due, due date, billing period, utility provider, and account or reference identifier. OCR is limited to document-information extraction and does not determine billing classification, confirm payment, or replace validation. Extracted billing information is checked for required fields, provider and account/reference mapping, billing period, duplicate status, and other defined validation conditions. Records that are incomplete, uncertain, duplicated, unmappable, or unable to

satisfy the defined rules are sent to the Exception Queue for Manager review and correction. Corrected records may be revalidated and reprocessed.

The system processes valid records according to the defined billing and computation rules. It uses three payment statuses: Unpaid, Tenant Reported Paid, and Paid. Tenant Reported Paid indicates a tenant-reported payment and does not represent confirmed payment. Only the Manager can confirm payment and change the status to Paid. A digital receipt is generated only after Manager-confirmed payment. The system does not include a payment gateway, bank integration, automatic payment verification, independent external payment verification, or direct utility-payment verification. The Manager performs any required payment verification before confirming payment in the system.

The system provides a tenant-facing dashboard for viewing recorded rental and utility dues, billing details, payment status, and confirmed-payment receipts. Tenant alerts use Web Push Notifications and direct tenants to the authenticated web portal. The system does not use SMS or email as tenant-notification channels. Web Push does not confirm payment, prove that payment occurred, prove that a tenant read the notification, or replace the billing record. The system may send permitted billing updates and due-date-related alerts, including a configured reminder three days before the due date. Notification delivery behavior is described and evaluated according to the implemented notification mechanism.

The system includes Admin, Manager, and Tenant roles. The Admin role is limited to system maintenanceThe Manager manages properties, units, tenants, utility mappings, billing rules, billing records, exceptions, corrections, payment confirmation, and receipts. The Tenant views billing information and payment status, reports payment, and receives permitted Web Push Notifications. Role-based access restricts functions and information according to the assigned role.

The system uses a centralized database for property, unit, tenant, utility-mapping, rent-configuration, billing, payment-status, exception, correction, receipt, and applicable notification records.

The system is evaluated using five selected ISO/IEC 25010:2023 quality characteristics: Functional Suitability, Performance Efficiency, Interaction Capability, Reliability, and Security. The evaluation is limited to these selected characteristics and does not represent an assessment of every characteristic in the ISO/IEC 25010:2023 product-quality model. The study evaluates the system according to its defined functions and approved testing procedures and does not claim to measure long-term changes in property revenue, tenant behavior, tenant payment compliance, business profitability, or overall operational performance unless these outcomes are specifically included in the evaluation.

Within its defined boundaries, the study does not include a mobile application; installation or physical reading of Internet of Things smart meters, electrical submeters, or water submeters; Internet or Wi-Fi billing; predictive profit analytics; predictive billing;

Artificial Intelligence; Machine Learning; Deep Learning; Generative AI; blockchain technology; automated utility disconnection; payment gateways; bank integration; independent external payment verification; maintenance-ticket management; legal contract generation; or automated eviction processing. The OCR component does not cover handwritten bills, heavily damaged documents, severely distorted documents, or low-resolution photographs. The study also does not claim that the system independently verifies real-world payments, guarantees notification delivery or reading, establishes a universally fair utility-allocation formula, or provides complete legal certification under the Philippine Data Privacy Act.

CHAPTER II

METHODOLOGY

This chapter presents the methodology and procedures used in the development of the proposed *Intelligent Property Management Workflow with OCR-Enabled Automated Billing Classification and Tenant Notification*. It describes the project requirements, system design, conceptual operation, interface mockups, and Iterative Software Development Methodology that guide the development and evaluation of the system.

2.0 Project Overview

Project Information

The project, entitled *Intelligent Property Management Workflow with OCR-Enabled Automated Billing Classification and Tenant Notification*, is a web-based property-management system designed for selected boarding houses and residential apartments in Pasig City. It covers the management of properties, units, tenants, rent, electricity, water, utility accounts or reference identifiers, billing records, payment statuses, exceptions, digital receipts, and tenant notifications.

The system provides a centralized workflow for receiving electronic utility bills, extracting billing information from supported documents through Optical Character

Recognition (OCR), validating extracted information, applying predefined rule-based billing conditions, computing applicable charges, managing exceptions, confirming payments, generating digital receipts, and triggering permitted Web Push Notifications. The designated property-management email account is used for utility-bill intake and attachment retrieval only. It is not used as a tenant-notification channel.

In this study, the term *intelligent* refers to predefined, explicit, deterministic, and traceable rule-based processing. It does not refer to Artificial Intelligence, Machine Learning, Deep Learning, Predictive Analytics, Generative AI, or any machine-learning model. The system follows an "automate by default, intervene by exception" approach. Records that satisfy the defined processing conditions are processed through the applicable rules, while records requiring judgment, correction, or further review are routed to the Exception Queue for Manager intervention.

The system supports three payment statuses: Unpaid, Tenant Reported Paid, and Paid. Tenant Reported Paid indicates that the tenant reports a payment but does not represent confirmed payment. Only the Manager can confirm payment and change the status to Paid. A digital receipt is generated only after Manager-confirmed payment. Tenant notifications use Web Push Notifications for permitted billing updates and due-date-related alerts, including a configured reminder three days before the due date. Web Push Notifications direct tenants to the authenticated web portal and do not confirm payment, prove that payment occurred, or prove that a tenant read the notification.

Requirements Definition

The proposed system is designed to meet the following business, user, and security requirements.

A. Business Requirements

The proposed system shall:

1. Receive electronic utility bills through a designated property-management email account and retrieve supported billing attachments.
2. Extract available billing information from supported standard, digitally generated PDF files and sufficiently clear image files using OCR.
3. Extract the following target fields when available:
    - Amount due.
    - Due date.
    - Billing period.
    - Utility provider.
    - Account or reference identifier.
4. Validate extracted billing information using predefined conditions, including:
    - Required-field checking.
    - Provider mapping.
    - Account or reference mapping.
    - Billing-period checking.



- Duplicate-status checking.
- Other defined validation conditions.

5. Map utility records to the appropriate property, unit, utility account or reference identifier, and provider using Manager-configured mappings.
6. Apply predefined, explicit, deterministic, and traceable rules for:
    - Billing validation.
    - Provider and account/reference mapping.
    - Duplicate detection.
    - Billing classification.
    - Individual or shared billing identification.
    - Billing computation according to configured rules.
7. Support Individual Billing Arrangement and Shared-Occupancy Billing Arrangement.
8. Generate monthly rent according to the tenant's configured billing or lease-anniversary date.
9. Apply fixed and non-prorated rent rules.
10. Record one-month advance rent as a separate line item in the first billing record.
11. Process billing categories limited to rent, electricity, and water.
12. Allocate applicable rent and utility charges among active tenants under the Shared-Occupancy Billing Arrangement according to the predefined allocation rule selected or configured by the Manager.

13. Record shared-billing allocations using exactly two decimal places and ensure that the total of the allocated shares equals the original source billing amount.
14. Process and finalize billing records according to the applicable predefined rules when all required validation, mapping, duplicate-checking, and billing conditions are satisfied.
15. Route records that do not satisfy the defined processing conditions to the Exception Queue for Manager review and correction.
16. Allow corrected exception records to be revalidated and reprocessed or recomputed when applicable.
17. Store property, unit, tenant, utility-mapping, rent-configuration, billing, payment-status, exception, correction, receipt, and applicable notification information in a centralized database.
18. Maintain the payment statuses Unpaid, Tenant Reported Paid, and Paid.
19. Distinguish tenant-reported payment from Manager-confirmed payment.
20. Allow only the Manager to confirm a tenant-reported payment and change the status to Paid.
21. Generate a digital receipt only after Manager-confirmed payment.
22. Provide a tenant-facing dashboard for viewing recorded rental and utility dues, billing details, payment status, and confirmed-payment receipts.
23. Trigger permitted Web Push Notifications for billing updates and due-date-related alerts, including a configured reminder three days before the due date.

24. Direct tenants through Web Push Notifications to the authenticated web portal where complete billing information is available.
25. Apply a defined correction process for exception and finalized records. For exception records, the process shall include Manager review and correction, revalidation, and reprocessing or recomputation when applicable. For finalized records, the process shall include Manager correction, revalidation, recomputation when applicable, and record update.
26. Maintain the distinction between routine automated processing and records requiring Manager intervention.

The system shall not use SMS, GSM, or tenant email notifications. The designated property-management email account shall be used only for utility-bill intake and attachment retrieval.

B. User Requirements

The proposed system includes three user roles: Admin, Manager, and Tenant.

Admin

The Admin role is limited to system maintenance. The Admin shall be able to perform only the system-maintenance functions included in the implemented system.

Manager

The Manager shall be able to:

1. Log in to the system using valid credentials.
2. Manage property information.
3. Manage unit information.
4. Manage tenant information.
5. Configure utility providers and account or reference mappings.
6. Configure rent information and billing arrangements.
7. Configure applicable billing rules.
8. View and manage billing records.
9. Review billing information extracted through OCR when necessary.
10. Review billing validation and classification results.
11. Review records placed in the Exception Queue.
12. Correct billing or payment information when Manager intervention is required.
13. Revalidate and reprocess or recompute corrected records when applicable.
14. Manage post-finalization corrections according to the defined correction process.
15. Confirm tenant-reported payments.
16. View payment status for applicable rent, electricity, and water billing records.
17. View confirmed-payment receipts.
18. Manage relevant Web Push Notification functions within the defined system scope.
19. Perform other Manager-only operations included in the implemented system.

The Manager may correct an exception before finalization or correct an existing billing or payment record after finalization. A corrected record shall undergo revalidation and recomputation when applicable.

Tenant

The Tenant shall be able to:

1. Log in to the authenticated system.
2. View assigned property or unit information.
3. View applicable rent, electricity, and water billing records.
4. View billing details, including billing category, billing period, and amount due.
5. View payment status.
6. Report an applicable payment through the system.
7. View digital receipts for Manager-confirmed payments.
8. Receive permitted Web Push Notifications.
9. Access the authenticated web portal through permitted Web Push billing and due-date-related alerts.

A tenant-reported payment shall not automatically change the payment status to Paid. The status shall remain Tenant Reported Paid until the Manager confirms the payment.

C. Security Requirements

The proposed system is designed to protect property, tenant, billing, payment, utility, exception, correction, receipt, and applicable notification information processed by the application.

The proposed system shall include the following security requirements.

User Authentication

Users shall provide valid credentials before accessing protected system functions.

Role-Based Access Control

System functions and information shall be restricted according to the assigned user role. Admin, Manager, and Tenant permissions shall follow their defined responsibilities.

Data Access Restriction

Property, tenant, billing, payment, utility, exception, correction, receipt, and applicable notification information shall be accessible only to authorized users and functions.

Secure Utility-Bill Intake

Electronic utility-bill documents retrieved through the designated property-management email workflow shall be accessible only through authorized system functions and to users with appropriate permissions.

Billing and Payment Information Protection

Billing and payment information shall be restricted from unauthorized modification or access through authentication, role-based access, and defined system permissions.

Session Security

Authenticated user sessions shall be managed according to the implemented access-control mechanism.

Input Validation

User-provided information and extracted billing information shall be validated before being stored or processed.

Controlled Exception Review

Billing records that fail required-field validation, identification, duplicate checking, mapping, or other defined processing conditions shall be routed to the Exception Queue for Manager review.

Correction Information

Correction information shall be stored within the centralized database as applicable.

The security requirements are limited to the defined system functions, access controls, and security-related criteria included in the study's evaluation procedures. They do not constitute a legal audit, certification, or claim of complete compliance with the Philippine Data Privacy Act.

2.1 Concept of the Study

Architectural Framework

The architectural framework presents the structure and interaction of the proposed Intelligent Property Management Workflow with OCR-Enabled Automated Billing Classification and Tenant Notification. It shows the major components, actors, and information relationships within the proposed system.

The framework organizes routine processing components and exception-handling components within the proposed system. The Exception Queue provides a controlled point for Manager review and correction of records that do not satisfy the defined processing conditions.

The framework consists of the following major components and actors:

- Property and Unit Management
- Tenant Management
- Utility Provider and Account/Reference Mapping
- Rent Configuration
- Property-Management Email Intake
- OCR Processing
- Validation and Rule-Based Billing Classification
- Duplicate Checking
- Billing Computation

- Exception Queue
- Billing Records
- Payment Status Management
- Digital Receipt
- Web Push Notification
- Centralized Database
- Admin
- Manager
- Tenant
- Utility Provider

Property and Unit Management manages property and unit information. Tenant Management manages tenant information and its association with the applicable property or unit. The Utility Provider and Account/Reference Mapping component manages utility providers, account or reference identifiers, and their configured property or unit relationships. The Manager configures the applicable mappings.

Rent Configuration manages the tenant's configured rent information and billing or lease-anniversary date. The Centralized Database stores the corresponding configuration records. The system applies the fixed monthly rent rule, does not prorate rent, and records one-month advance rent as a separate line item in the first billing record.



Property-Management Email Intake receives supported electronic utility-bill attachments through the designated property-management email account. The email account is used only for utility-bill intake and attachment retrieval and is not used as a tenant-notification channel. OCR Processing extracts available billing information from supported documents, including the amount due, due date, billing period, utility provider, and account or reference identifier.

Validation and Rule-Based Billing Classification checks extracted information using predefined rules. It validates required fields, identifies the applicable property and unit using the configured mapping, checks the billing period and duplicate status, and determines whether the applicable billing arrangement is Individual Billing Arrangement or Shared-Occupancy Billing Arrangement. The rules are predefined, explicit, deterministic, and traceable. The system does not use Artificial Intelligence, Machine Learning, Deep Learning, Predictive Analytics, Generative AI, or any machine-learning model for this processing.

Billing Computation calculates applicable rent and utility charges according to the configured rules. Under the Shared-Occupancy Billing Arrangement, applicable charges are allocated among active tenants using the predefined allocation rule selected or configured by the Manager. Allocated amounts use exactly two decimal places, and the total of the allocated shares must equal the original source billing amount.

The Exception Queue manages records requiring Manager intervention. It receives records that are invalid, incomplete, uncertain, duplicated, unmappable, or

otherwise unable to satisfy the defined processing conditions. The Manager reviews and corrects these records. Corrected information is subject to revalidation and reprocessing or recomputation when applicable.

Billing Records manages applicable billing information and finalized billing records. Payment Status Management manages the statuses Unpaid, Tenant Reported Paid, and Paid for applicable rent, electricity, and water billing records. Tenant Reported Paid represents a tenant-reported payment and is not considered confirmed until the Manager reviews and confirms the payment. The Digital Receipt component generates a receipt only after Manager-confirmed payment.

The Centralized Database stores applicable property, unit, tenant, utility-mapping, rent-configuration, billing, payment-status, exception, correction, receipt, and notification information. Correction information is stored within centralized records rather than as a separate architectural component.

The Web Push Notification component provides permitted billing updates and due-date-related alerts, including the configured reminder three days before the due date. It directs tenants to the authenticated web portal and does not confirm payment, prove that payment occurred, prove that a tenant read the notification, or replace the billing record.

The Admin performs system-level maintenance. The Manager manages property information, tenant records, utility mappings, billing configurations, billing records,

exceptions, corrections, payment confirmations, receipts, and relevant Web Push Notification functions. The Tenant views billing information and payment status, reports applicable payments, views confirmed-payment receipts, and receives permitted Web Push Notifications. The Utility Provider serves as the source of electronic utility-bill documents received through the designated property-management email account.

The components operate as an integrated web-based property-management system. Property and tenant management, utility mapping, and rent configuration provide the required management information. Billing-processing components support utility-bill intake, OCR extraction, validation, rule-based classification, duplicate checking, billing computation, and exception handling. Billing and payment components manage billing records, payment statuses, and confirmed-payment receipts. The Web Push Notification component supports permitted billing and due-date-related alerts, while the Centralized Database stores the applicable system records. The Admin, Manager, Tenant, and Utility Provider interact with the system according to their defined roles and responsibilities.

2.2 Conceptual Operation

The proposed system operates through a defined workflow covering rent, electricity, and water. Rent billing follows the tenant's configured billing or lease-anniversary date, fixed monthly rent amount, non-prorated billing rule, and one-month advance-rent requirement for the first billing record.

For utility billing, the system receives electronic utility-bill documents through the designated property-management email account. Supported attachments are retrieved and processed using OCR. The extracted information is validated and processed through predefined rules for mapping, duplicate checking, billing classification, and computation.

Valid records proceed to billing finalization. Records that are invalid, incomplete, uncertain, duplicated, unmappable, or otherwise unable to satisfy the defined processing conditions are placed in the Exception Queue for Manager review and correction. Corrected records may be revalidated, reprocessed, or recomputed when applicable.

The system manages the payment statuses Unpaid, Tenant Reported Paid, and Paid. A finalized billing record begins with the applicable unpaid status. When a Tenant reports payment, the status changes to Tenant Reported Paid. The status remains unconfirmed until the Manager confirms the payment. After confirmation, the status changes to Paid and a digital receipt is generated.

The system may trigger permitted Web Push Notifications for billing updates and due-date-related alerts, including the configured reminder three days before the due date. Web Push Notifications direct Tenants to the authenticated web portal. They do not confirm payment, prove that payment occurred, or prove that a Tenant read the notification.

Process Flowchart

The process flowchart illustrates the sequence and decision points of the proposed billing workflow. It includes the separate processing paths for rent and utility bills, OCR extraction, validation, mapping, duplicate checking, billing classification, billing computation, exception handling, billing finalization, payment confirmation, digital receipt generation, and permitted Web Push Notifications.

For rent billing, the system checks the tenant's configured billing or lease-anniversary date and determines whether the billing is the first billing. If it is the first billing, the system records one-month advance rent as a separate line item. The system then applies the configured fixed and non-prorated rent rules.

For utility billing, the process begins when the Utility Provider sends an electronic utility bill to the designated property-management email account. The system retrieves the supported attachment and uses OCR to extract available billing information. The extracted information is checked for required fields, provider and account/reference mapping, billing period, duplicate status, and other defined validation conditions.

If the record satisfies the defined conditions, it proceeds to billing computation and finalization. If the record is invalid, incomplete, uncertain, duplicated, unmappable, or otherwise unable to satisfy the defined processing conditions, it is placed in the

Exception Queue. The Manager reviews and corrects the record before it is revalidated and reprocessed or recomputed when applicable.

After a billing record is finalized, its payment status is initialized as Unpaid. When the Tenant reports a payment, the status changes to Tenant Reported Paid. Only the Manager can confirm the payment and change the status to Paid. A digital receipt is generated only after the status becomes Paid.

The flowchart may show a permitted Web Push Notification after applicable billing updates or for the configured reminder three days before the due date. The notification directs the Tenant to the authenticated web portal. The flowchart must not show SMS, tenant email notifications, notification retries, second-failure alerts, or notification read confirmation unless those functions are verified as implemented.

Full-Blown Data Flow Diagram

The Full-Blown Data Flow Diagram illustrates the movement of billing, payment, exception, correction, receipt, and applicable notification information among the external actors, system processes, and centralized database.

The process begins when the Utility Provider sends an electronic utility bill to the designated property-management email account. Property-Management Email Intake retrieves supported attachments and sends them to OCR Processing. OCR Processing extracts available billing information, including the amount due, due date, billing period, utility provider, and account or reference identifier.

The extracted information is sent to Validation and Rule-Based Billing Classification. This process checks required fields, identifies the applicable property or unit through configured mappings, checks the billing period and duplicate status, and determines the applicable billing arrangement. Valid information proceeds to Billing Computation, while invalid, incomplete, uncertain, duplicated, unmappable, or otherwise unprocessable information is sent to the Exception Queue.

The Manager receives exception information and may correct affected records. Corrected exception records are revalidated and reprocessed or recomputed when applicable. For post-finalization corrections, the Manager corrects the affected record, after which the system revalidates, recomputes when applicable, and updates the record.

Billing Computation processes rent, electricity, and water according to the configured rules. Rent follows the configured billing or lease-anniversary date, fixed monthly amount, non-prorated rule, and first-billing advance-rent requirement. Shared-occupancy charges are allocated among active tenants according to the predefined allocation rule selected or configured by the Manager. Allocated amounts use two decimal places, and their total must equal the original source billing amount.

Finalized billing information is stored in the Centralized Database. Payment Status Management maintains Unpaid, Tenant Reported Paid, and Paid statuses. The Tenant may report a payment, but only the Manager can confirm the payment. After confirmation, the status becomes Paid and Digital Receipt generates the applicable receipt.

The Web Push Notification function triggers permitted billing updates and due-date-related alerts for the Tenant. The notification directs the Tenant to the authenticated web portal. The DFD must not include SMS, tenant email notification, notification retry, notification-failure alerts, payment gateways, bank integration, or external payment-verification processes unless such functions are verified as implemented.

The Admin is limited to system-level maintenance. The Manager manages configured system information, billing records, exceptions, corrections, payment confirmation, receipts, and relevant notification functions. The Tenant views billing information and payment status, reports payment, views confirmed-payment receipts, and receives permitted Web Push Notifications.

2.3 Mockup Design

The mockups present the proposed interface and interaction flow of the web-based property-management workflow. They illustrate the expected arrangement of system functions and information for the Admin, Manager, and Tenant roles. The mockups serve as visual references for the development of the system interface.

Admin Interface

The Admin interface shall show only system-level maintenance functions that are included in the implemented system. It shall not include ordinary property-management, billing, payment-confirmation, exception-review, or Tenant functions unless the system explicitly assigns those functions to the Admin role.



VERIFY: The specific Admin maintenance functions must match the actual implementation and mockups.

Manager Interface

The Manager mockups may include the following implemented functions:

- Manager login
- Manager dashboard
- Property, unit, and tenant management
- Billing and rent configuration
- Utility-provider and account/reference mapping
- Incoming utility bills and OCR review
- Billing validation and classification review
- Exception Queue

- Billing records and payment status

- Payment confirmation

- Confirmed-payment receipts

- Correction handling

- Relevant Web Push Notification functions

The Manager mockups must not include SMS settings, GSM controls, tenant-email notification settings, notification-retry controls, or unsupported notification-failure dashboards.

Tenant Interface

The Tenant mockups may include:

- Tenant login

- Tenant dashboard

- Assigned property or unit information

- Billing details

- Billing category

- Billing period

- Amount due

- Payment status

- Report Payment

- Confirmed-payment digital receipt

- Tenant Web Push Notifications

The Tenant interface must not imply that Tenant Reported Paid is confirmed payment. It must not display SMS or tenant-email notification preferences, notification-read proof, or payment-verification controls.

Use the following labels for the mockups:

- Login Portal

- Admin Dashboard

- Manager Dashboard

- Property, Unit, and Tenant Management

- Billing Configuration

- Incoming Bills and OCR Review

- Exception Queue

- Billing Records and Payment Status

- Notifications and Receipts

- Tenant Dashboard

- Tenant Billing Details

- Tenant Report Payment

- Confirmed-Payment Digital Receipt

- Tenant Notifications

Replace “Tenant Notifications and Preferences” with “Tenant Notifications” unless notification preferences are verified as part of the implementation.

2.4 Methodology of the Study

The study will use the Iterative Software Development Methodology to develop the proposed *Intelligent Property Management Workflow with OCR-Enabled Automated Billing Classification and Tenant Notification*. The methodology will organize development into successive iterations in which related functional areas will be developed, reviewed, refined, and integrated into the overall system.

This approach will allow the researchers to address the system requirements in manageable groups while maintaining consistency among the system’s property-management, billing, payment, exception, receipt, database, and Web Push Notification functions. The methodology will support an “automate by default, intervene by exception” approach in which valid records proceed through defined rules and records requiring review are directed to the Exception Queue.

The development will begin with the establishment of the system requirements, user roles, and core data structure. It will continue with property and billing configuration, property-management email intake and OCR, rule-based classification and exception handling, billing computation and record management, payment and receipt functions, Web Push Notifications, and system-wide integration and testing. Each iteration will focus on a specific group of related functions while considering its connection to previously developed components. The methodology will support continuous review and refinement before the final system evaluation.

2.4.1 Principles of the Methodology

The Iterative Software Development Methodology will be applied through the following principles.

Incremental Development

The system will be developed through smaller functional iterations rather than through one complete implementation. Each iteration will focus on a related group of requirements, allowing the researchers to build the system progressively.

Continuous Testing and Review

Testing and review will be conducted during development to identify issues in the functions included in each iteration. The results of testing and review will be used as a basis for necessary corrections and improvements. No completed testing result is claimed in this chapter.

Refinement Between Iterations

Issues and requirements identified during an iteration will be reviewed before related functions are integrated into succeeding iterations. This will allow the researchers to refine the system while development progresses.

Cumulative Integration

Functions developed in earlier iterations will be integrated with functions developed in later iterations. This will allow the researchers to gradually form a complete system while checking the interaction among its functional areas.

Exception-Based Intervention

The system will automate valid records according to predefined rules while directing invalid, incomplete, uncertain, duplicated, unmappable, or otherwise unprocessable records to the Exception Queue for Manager review. This principle does not mean that every record will be processed without human intervention.

2.4.2 Development Iterations

The development of the proposed system will be organized into seven planned iterations based on the functional requirements of the study. The iterations describe the planned development structure and do not claim that the iterations have already been completed.

Table 2.1

Development Iterations of the Proposed System

| Iteration | Development Focus | Main Functions |

|---|---|---|

| Iteration 1 | Requirements and System Foundation | User roles, authentication, property, unit, tenant, and core centralized database structure |

| Iteration 2 | Property and Billing Configuration | Utility providers, account or reference mapping, rent configuration, billing arrangements, and billing rules |

| Iteration 3 | Property-Management Email Intake and OCR | Designated property-management email intake, attachment retrieval, supported document processing, and OCR extraction |

| Iteration 4 | Rule-Based Classification and Exception Handling | Validation, property or unit identification, duplicate checking, billing classification, Exception Queue, Manager review, correction, and revalidation |

| Iteration 5 | Billing Computation and Records | Rent, electricity, and water computation; individual and shared-occupancy billing; billing records; and finalization |

| Iteration 6 | Payment, Receipt, and Web Push Notification | Payment statuses, Tenant payment reporting, Manager confirmation, digital receipts, permitted Web Push billing updates, and the configured due-date reminder three days before the due date |

| Iteration 7 | System Integration and Testing | Integration of system functions, role-based workflows, security controls actually implemented, system-wide testing, and final evaluation |

2.4.3 Planned Application of Each Iteration

Iteration 1 – Requirements and System Foundation

This iteration will establish the basic structure of the proposed system based on the identified requirements. It will cover the Admin, Manager, and Tenant roles; user authentication; property and unit information; tenant records; and the initial centralized database structure. The purpose of this iteration will be to establish the foundation required by the succeeding system functions.

Iteration 2 – Property and Billing Configuration

This iteration will focus on the configurations required for billing operations. It will include utility-provider information, account or reference identifiers, property and unit mappings, rent configuration, billing arrangements, and applicable billing rules.

Rent configuration will follow the tenant's configured billing or lease-anniversary date. Rent will be fixed and non-prorated. One-month advance rent will be configured as a separate line item for the first billing record.

The Manager will select or configure the predefined allocation rule for Shared-Occupancy Billing Arrangement. The allocation formula, rounding procedure, and reconciliation method will be treated as system-design decisions and evaluated during testing.

Iteration 3 – Property-Management Email Intake and OCR

This iteration will focus on receiving electronic utility bills through the designated property-management email account. Supported attachments will be retrieved and

processed individually. OCR will be used to extract available billing information from supported standard, digitally generated PDF files and sufficiently clear image files.

The target fields will include the amount due, due date, billing period, utility provider, and account or reference identifier. OCR will function only as a document-information-extraction mechanism. It will not classify billing records, make billing decisions, confirm payments, or replace validation.

Iteration 4 – Rule-Based Classification and Exception Handling

This iteration will focus on applying predefined, explicit, deterministic, and traceable rules to the extracted and configured billing information. The system will validate required fields, identify the applicable property or unit using configured mappings, check the billing period and duplicate status, and determine the applicable billing arrangement.

Records that are invalid, incomplete, uncertain, duplicated, unmappable, or otherwise unable to satisfy the defined processing conditions will be routed to the Exception Queue for Manager review and correction. Corrected records will undergo

revalidation before continuing through the billing process. Reprocessing or recomputation will occur when applicable.

Iteration 5 – Billing Computation and Records

This iteration will focus on the computation and management of billing records for rent, electricity, and water. Individual and Shared-Occupancy Billing Arrangements will be supported using the configured billing rules.

For Shared-Occupancy Billing Arrangement, applicable rent and utility charges will be allocated among active tenants using the predefined allocation rule selected or configured by the Manager. Allocated amounts will use exactly two decimal places, and the total of the allocated shares will equal the original source billing amount. Valid records will proceed to billing finalization and storage.

Iteration 6 – Payment, Receipt, and Web Push Notification

This iteration will focus on payment-status management, digital receipts, and permitted tenant-notification functions. Payment statuses will include Unpaid, Tenant Reported Paid, and Paid. A tenant-reported payment will remain unconfirmed until the Manager confirms the payment. A digital receipt will be generated only after Manager-confirmed payment.

Tenant notifications will use Web Push Notifications for permitted billing updates and the configured reminder three days before the due date. Web Push Notifications will direct Tenants to the authenticated web portal. They will not confirm payment, prove that payment occurred, or prove that a Tenant read the notification.

Iteration 7 – System Integration and Testing

This iteration will focus on integrating the planned system functions into a complete workflow and conducting system-wide testing. The researchers will check the interaction among user roles, property and tenant management, utility mapping, rent configuration, billing processing, payment-status management, exception handling, corrections, receipts, Web Push Notifications, and centralized database functions.

The system will be evaluated using the selected ISO/IEC 25010:2023 quality characteristics:

- Functional Suitability

- Performance Efficiency

- Interaction Capability

- Reliability

- Security

The final evaluation will be reported based on the approved testing and evaluation procedures. No evaluation result is claimed in this chapter.

2.4.4 Application of the Methodology to the Study

The Iterative Software Development Methodology will provide a structured approach for developing the proposed system according to its functional requirements. The system will be developed progressively, beginning with the core data and user

functions and continuing with billing configuration, property-management email intake, OCR, rule-based processing, exception handling, billing computation, payment, receipts, Web Push Notifications, and system integration.

Each iteration will be reviewed before succeeding functions are integrated. The review process will help identify inconsistencies between requirements, data structures, workflows, and user functions.

The methodology will also support the controlled handling of system exceptions. Instead of requiring all billing records to proceed automatically, records that do not satisfy the defined processing conditions will be directed to the Exception Queue for Manager review and correction. This will allow the researchers to refine the processing rules and related functions while maintaining the intended “automate by default, intervene by exception” approach.

For exception records, the correction process will include Manager review and correction, revalidation, and reprocessing or recomputation when applicable. For finalized records, the correction process will include Manager correction, revalidation, recomputation when applicable, and record update.

Through the planned iterations, the functional areas of the proposed system will be integrated into a complete workflow. Testing will be conducted during development and system integration to identify defects and verify whether the implemented functions meet the defined requirements. The completed system will then undergo final evaluation using the selected ISO/IEC 25010:2023 quality characteristics. No completed iteration, testing result, or evaluation result is claimed in this chapter.